Literature Resolution of the Translation-Invariant Min–Max Question

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Status. The selected question from Guillen–Schwab's 2016 paper is answered affirmatively, and in the exact requested form, by their later paper on Euclidean space. This packet records the match and does not claim a new proof.

The source question

The 2016 paper proves min–max representations for nonlinear operators satisfying the global comparison property. In its concluding list of questions, the authors ask whether, when the underlying manifold is $\mathbb{R}^d$ and the nonlinear operator I is translation invariant, the linear operators in the min–max representation can all be chosen translation invariant [1, Section 1.5].

Here translation invariance means that, for translations $\tau_x u(y) = u(x+y)$,

$$I(\tau_x u, y) = I(u, x+y).$$

Equivalently, the whole operator is recovered from the scalar functional $F(u) = I(u, 0)$ by $I(u, x) = F(\tau_x u)$.

The later theorem

The later paper states the following result as Theorem 1.10 [2, Theorem 1.10].

Theorem 1 (Guillen–Schwab)**.** *Suppose*

$$I : C_b^2(\mathbb{R}^d) \longrightarrow C_b^0(\mathbb{R}^d)$$

is Lipschitz, has the global comparison property, and is translation invariant. Then there are index sets and constants f_{ab}, together with linear translation-invariant constant-coefficient Lévy operators L_{ab}, such that

$$I(u, x) = \min_a \max_b \{ f_{ab} + L_{ab}(u, x) \}.$$

Every linear operator appearing in the representation therefore commutes with translations.

This is precisely the affirmative conclusion requested in the selected source question. The theorem imposes the natural Euclidean function-space and Lipschitz hypotheses under which the original min–max theory is formulated.

Why the proof resolves the structural issue

The later proof does more than infer translation invariance indirectly. It starts with the scalar functional

$$F(u) = I(u, 0),$$

applies the scalar min–max representation to F, and identifies its supporting linear functionals with constant-coefficient Lévy operators. Translation invariance then supplies

$$I(u, x) = F(\tau_x u),$$

so the same constant-coefficient operators give the representation at every point x. Thus the representing family is translation invariant by construction, exactly addressing the possible loss of symmetry raised in 2016 [2, proof of Theorem 1.10].

Scope of this resolution

The 2016 article ends with several broader questions concerning manifolds, regularity, and alternative domains. The literature theorem above resolves the selected Euclidean translation-invariance question only. No claim is made here about the other questions in that list.

Verification record

The source question was checked in the authors' 2016 arXiv version. The statement and proof mechanism above were checked in arXiv:1812.09642v2, where the result is numbered Theorem 1.10. The later paper was published in *Nonlinear Analysis* 193 (2020), article 111468, DOI 10.1016/j.na.2019.02.021.

References

[1] N. Guillen and R. W. Schwab, *Min–max formulas for nonlocal elliptic operators*, arXiv:1606.08417 (2016), https://arxiv.org/abs/1606.08417.

[2] N. Guillen and R. W. Schwab, *Min–max formulas for nonlocal elliptic operators on Euclidean space*, Nonlinear Analysis 193 (2020), 111468; arXiv:1812.09642v2, https://arxiv.org/abs/1812.09642.